Routes you can read, choose and save

Python + GraphHopper REST API

PalpakGPT

Activities 3 and 4 | October 18, 2026

What problem are we solving?

Choose the intended place

Several places may share the same name. The user needs enough detail to choose.

Read a useful route

Show familiar units, clear totals and directions without asking users to read JSON.

Keep the result

Save directions locally and reuse a route during the same session.

Build on the lab

Retain the REST API workflow while improving input, output and error handling.

From terminal input to a reusable route

1 / Choose options

Travel profile, metric/imperial units, summary/detailed output.

2 / Geocode origin

Validate up to five candidates; show coordinates and accept the selected place.

3 / Geocode destination

Choose the destination; stop safely if a location is not found.

4 / Request routing

Send the selected coordinates and profile; validate the JSON path.

5 / Display and remember

Print totals/directions; retain the last ten successful routes.

6 / Export locally

TXT, JSON or CSV; check filenames and avoid silent overwrites.

Core enhancements: F1 to F4

F1 / Units

Metric or imperial values apply to totals and every direction step.

F2 / Clear output

Aligned summary, HH:MM:SS duration and numbered plain-text directions.

F3 / Output mode

Summary retains totals; detailed mode also shows directions.

F4 / Response checks

Validate fields and numeric ranges; recover safely and redact the key.

F7 / Save a route. F8 / Choose a place.

SYNTHETIC OFFLINE DEMO

```
1. Central Park, Example City (lat: 10.0, lng: 20.0)
2. Central Park, Example City (lat: 11.0, lng: 21.0)
Select a location: 2

Distance    : 2.0 miles
Duration    : 00:10:00
1. Turn left onto Example Street (1.0 miles)
2. Arrive at Demo Destination (1.0 miles)

Export formats: TXT | JSON | CSV
```

Menu and session history

Find a Route

Run the profile, unit, output-mode and location-selection workflow.

View Route History

Inspect any of the last ten successful routes without another HTTP request.

Export Previous Route

Save an existing session record in another supported format.

Session boundary

History is in memory. It disappears when the application exits.

Verification we can reproduce

69

offline unittest cases

2

CI Python versions: 3.10 / 3.14

0

live API calls in the offline suite

Merged-main CI passed; Jenkins execution is still required.

Live evidence is a separate record

Car

11.6 km | 1,079,659 ms | 10 direction steps

Bike

12.4 km | 2,532,665 ms | 39 direction steps

Foot

11.6 km | 8,585,308 ms | 72 direction steps

Disconnected route

Manila to Honolulu: HTTP 400, connection between locations not found.

Collaboration and review evidence

Scoped integration

F1/F2/F4/F3: PRs #21–#24. F9 pipeline code: PR #26.

F7/F8 batch

PR #27 merged October 18; exports, candidate selection, menu and history.

Review fixes

Show coordinates for identical labels; remove nine tracked cache files.

Attribution

Confirm each member's real contribution. Upload identity is not sole authorship.

Activity 4: Jenkins is required

Pipeline stages

Checkout → Install dependencies → Offline tests → Archive report.

Automatic detection

SCM polling supports local Jenkins; a webhook needs a reachable server.

Prepared setup

Local Docker/Compose package, Python/venv and required Jenkins plugins.

Still to verify

Real success, deliberate failure, recovery and new-commit trigger evidence.

Limits and the final team handoff

Provider limits

Coverage, quotas, profile access and availability depend on the API/account.

Application limits

Terminal UI; no map/database; only ten in-memory session records.

Team confirmation

Feature owners, actual contributions, scope approval and the presentation rehearsal.

Assessment still open

Complete the official rubric and personal reflections from actual experience.

Evidence to open during questions

Merged F7/F8 changes / PR #27

https://github.com/FlimsyOwl12/DEVASC-Project/pull/27

Merged-main 69-test CI run

https://github.com/FlimsyOwl12/DEVASC-Project/actions/runs/39121848255

Repository and task tracker

https://github.com/FlimsyOwl12/DEVASC-Project

Original lab reference

https://github.com/Cocoasaur/devasc-graphhopper-api/tree/2289d4d